Can SpaceX Pull Off a Buy Now Pay Later Gambit? Financing a $271B AI Buildout in 2027 - Multi-Channel

SemiAnalysis | Daniel Nishball, Oliver Kennon, Terence Ong | Monday, October 12, 2026

Executive Summary

- We expect SpaceX to add 5.2GW of AI capacity in 2027, requiring ~$271B of full project capex. SpaceX is reportedly raising $40B to buy Nvidia chips, but before any new financing the build leaves a $174B funding requirement through the end of 2027. We expect SpaceX to raise much more debt beyond the recently reported $40B bond and loan package.
- This $174B funding gap is huge but short: at current rental rates, a year of revenue pays for ~93-94% of the project cost of a Blackwell or Rubin cluster.
- SpaceX has a variety of tools at its disposal – the more traditional secured revolver, unsecured bonds, GPU financing and the unconventional Nvidia supplier credit – effectively Buy Now Pay Later (BNPL) – and customer prepayments.
- BNPL is a win-win for SpaceX and Nvidia: Nvidia already gives some investment grade customers 90 day to one-year credit terms, and extending those terms to one of its largest and fastest-growing customers can be a strategic move.
- Our view is that SpaceX will mix the tools. From 4Q F12/26 through 4Q F12/27, $343B of investing is covered by:
  - $108.7B of operating cash flow
  - $133.5B of existing cash, unsecured bond and loan issuance, Nvidia equity and power generation equipment financing which are highly likely
  - $40B drawn on a secured revolver (Tool A)
  - $20.1B net from Nvidia BNPL (Tool B)
  - $20B of 18-month unsecured bank loan (Tool C)
  - $10.4B of unsecured bonds (Tool C)
  - $10.3B net from customer prepayments (Tool E)
- The amounts above take care of 2027. If SpaceX adds another 5GW+ of capacity in 2028, we estimate it would need ~$150B or more of additional funding in 2028 from the same tools.

AI infrastructure is bringing a tidal wave of debt issuance. Our Compute Capital and Markets Model puts cumulative debt funding needs for AI servers and datacenters at $7T through 2029. Analyzing SpaceX shows one example of how that funding demand reaches the bond market in a huge way.

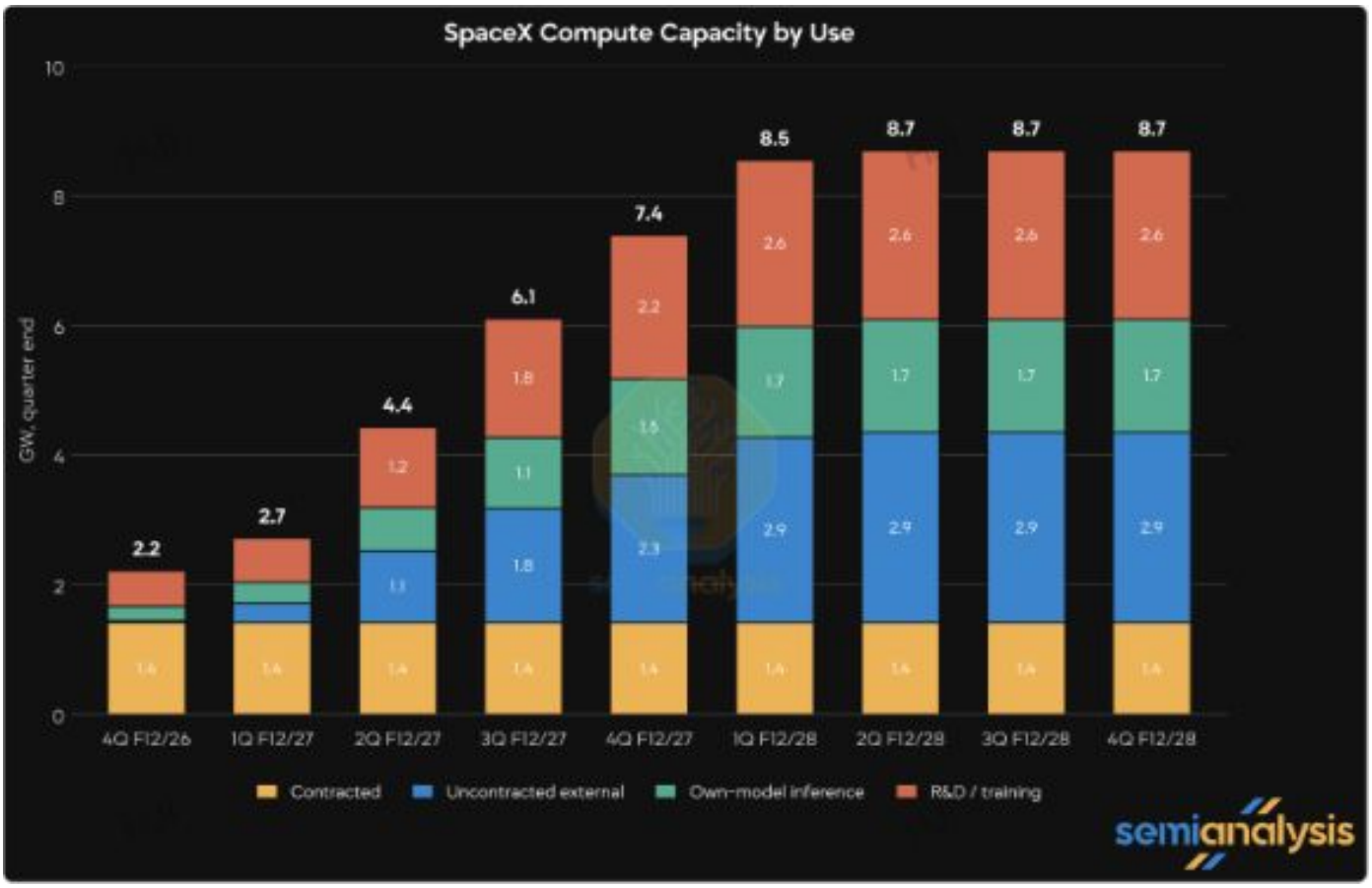
SpaceX raised $25B of investment grade senior notes in June and is reportedly seeking another $40B to buy Nvidia chips. We expect ultimate financing requirements to extend beyond the $40B proposed package, with more issuance from SpaceX in the next few quarters.

SpaceX should start 2027 with 2.2 GW of GPU-populated capacity. Our site-by-site financial model follows the latest published AI Datacenter Model schedule and reaches 7.4 GW by the end of 2027, comprising 7.14 GW of self-built capacity and 0.25 GW of leased capacity. We expect SpaceX to add 5.2 GW during 2027. For scale, Google and Meta had 7.8 GW and 6 GW of global datacenter capacity in 2025.

The 5.2 GW of additions represent roughly $271B of full project capex, inclusive of IT and datacenter capex. Payments start before commissioning, while 2027 cash capex also includes payments for later completions. The build consumes cash before it produces receipts. With no new financing or special supplier credit, there is a $174.0B funding requirement through 4Q F12/27 (the fourth quarter of the fiscal year ending December 2027)

subject to a $25B minimum cash balance.

Today, we focus specifically on how SpaceX can fund the 2027 build out, but we expect capacity to expand in 2028 as well, perhaps by 5-8GW – generating further funding requirements in 2028. We will analyze this separately when we present a more in-depth report on SpaceX and its multi-year operational and financial outlook.



Source: SemiAnalysis AI Datacenter Model, SemiAnalysis Compute Capital and Markets Model

How will this 2027 funding requirement be addressed? In this note we explore various options SpaceX has to plug this funding gap. We think that the use of the proposed $40B chip financing package, $25B Nvidia equity investment and equipment financing for half of on-site power generation capex is a given, and it forms our base case, leaving $105.4B more to be funded. SpaceX should be able to fill the rest with customer prepayments, traditional secured debt, unsecured bonds, GPU financing and the lever we think the market underappreciates the most: Nvidia's "Buy Now Pay Later" supplier credit.

What's on the Menu for 2027

We forecast compute revenue by analyzing three separate buckets:

- Contracted External Compute
- xAI's own inference business including R&D and training usage, and
- Uncontracted external capacity.

Contracted external compute is driven by five contracts with different delivery conditions and exit rights as set forth in the table below:

SpaceX Compute Agreements and Forecast Assumptions

Agreement, fees and hardware	Conditions and exit	Forecast treatment
Anthropic PBC $1.25B/month ~325k NVIDIA GPUs	Through May 2029. 90-day exit after three months.	496 MW modeled. Continuation and collection through 2028 assumed.
Google LLC $920M/month from Oct 2026 ~110k NVIDIA GPUs	Through Jun 2029. 90-day exit after Dec 2026. Delivery conditions apply.	232 MW modeled. Continuation and collection through 2028 assumed.
Reflection AI $150M/month from Jul 2026 GB300, as reported	Reported through 2029. 90-day exit after three months.	40 MW modeled. Continuation and collection through 2028 assumed.
July CSA Customer undisclosed $6.7B over six months from Oct 2026	Specific exit and renewal terms undisclosed.	343 MW modeled. $1.35B/month renewal and ramp phasing assumed.
September hosting Customer undisclosed $1.11B/month from Dec 2026	CFO portfolio remarks: 90-day commitment and exit. September terms undisclosed.	300 MW modeled. Continuation and collection through 2028 assumed.

MW, continued billing and renewals are forecast assumptions. Scheduled terms do not guarantee receipts.
Only Anthropic and Google legal obligors are verified. Other precise legal entities remain unverified.
4Q26 scheduled fees $10.77B; modeled revenue $9.92B after host-site gating.
July CSA: $2.70B scheduled, $2.10B modeled in 4Q26. 2027 agreement revenue $57.31B.

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Sources: Anthropic and Google SEC filings; reported Reflection terms; CFO remarks on the July CSA and September hosting deal. MW allocations, continued billing and collections are model assumptions.

The model allocates 1.4 GW to these agreements and assumes they continue through 2028, generating $14.3B per quarter from 1Q F12/27, or about $40.7M per MW annually. They contribute about 51% of F12/27 (the fiscal year ending December 2027) compute revenue. That share falls to about 35% on the 4Q F12/27 run-rate, as new billing capacity grows the other buckets. The six-month CSA schedules $2.7B in 4Q F12/26 and $4.0B in 1Q F12/27. The revised Southaven ramp limits recognized CSA revenue to $2.1B in 4Q F12/26 and total contract revenue to $9.9B, against $10.8B scheduled. The reserved capacity is not resold to other customers. Full billing resumes in 1Q F12/27, and the assumed CSA renewal then rises to $4.05B per quarter, or $16.2B annually.

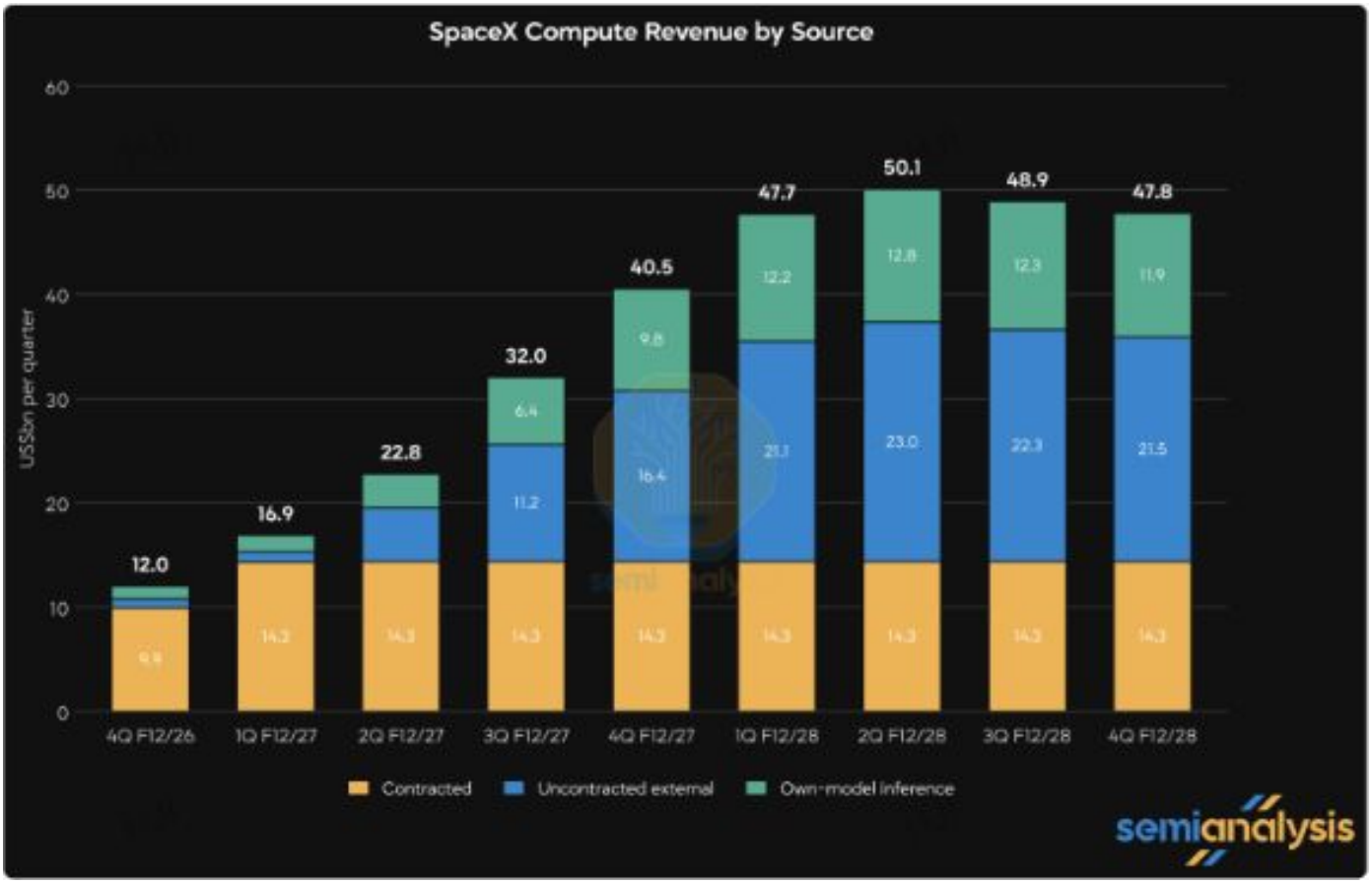
Turning to xAI's own inference and model business, we allocate 3.7 GW to such internal use at the end of 2027: 2.2 GW for R&D and frontier model training, with no modeled revenue, and 1.5 GW for serving xAI's own models to paying customers. We assume this inference capacity generates revenue per MW equal to 70% of the benchmark for renting GPUs to external customers.

Own-model inference generates a modeled $21.0B of revenue in 2027 and $49.2B in 2028. R&D and frontier training have no direct modeled revenue. The 70% realization factor assumes xAI can turn its inference capacity into paid demand.

Finally, for the uncontracted external bucket, we allocate 2.3 GW at the end of 2027 and assume 75% sell-through. After that adjustment, this capacity generates $33.9B of modeled revenue in 2027 and $87.9B in 2028. Each new Rubin cohort earns $50M per MW in year one, declining 15% annually to $42.5M in year two

and $26.1M in year five.

Uncontracted external compute represents 31% of year-end capacity and 30% of modeled 2027 compute revenue. On the 4Q F12/27 run-rate, it reaches about 40% of compute revenue. Most of the currently uncontracted external compute will be monetized via short-term contracts, which are priced at a significant premium to the longer-term five to six-year contracts that are more common in the market. This concentration makes rental pricing and paid utilization central to the credit case.



Source: SemiAnalysis Compute Capital and Markets Model

The funding gap is a timing problem

Our rental price curve assumes that newly available compute earns a high initial rate before prices decline. We use $40M per MW annually for new Blackwell cohorts, corresponding to $9.76/hr/GPU, and $50M per MW annually for Rubin, equivalent to $18.8/hr/GPU. Our SemiAnalysis GPU Pricing Index covers various tenors for all major GPUs including GB300s and VR NVL72s.

These rental rates are so high that gross annual revenue would equal about 94% of the representative $42.4M/MW Blackwell project capital cost and 93% of the $54.0M/MW Rubin capital cost, meaning the cluster can be completely paid for in just over a year. The funding gap is huge, but it is short in duration.

This fast paydown is also why annual capex figures are a poor guide to how much SpaceX must raise at any one point. What matters is how much has been paid for the next campus while the previous one is still ramping revenue. An operating fleet and a starting cash balance reduce the funding gap, but they do not eliminate the need for external capital during a build of this size.

Funding the 2027 buildout: $237.6B is in the bag, leaving $105.4B to raise

Let’s walk through what cash flow looks like for SpaceX quarter by quarter and how brisk operating cash flow generation, the $40B unsecured issuance, a $25B Nvidia equity investment and $8.4B of power generation leasing play out over time to cover some of 2027’s funding needs:

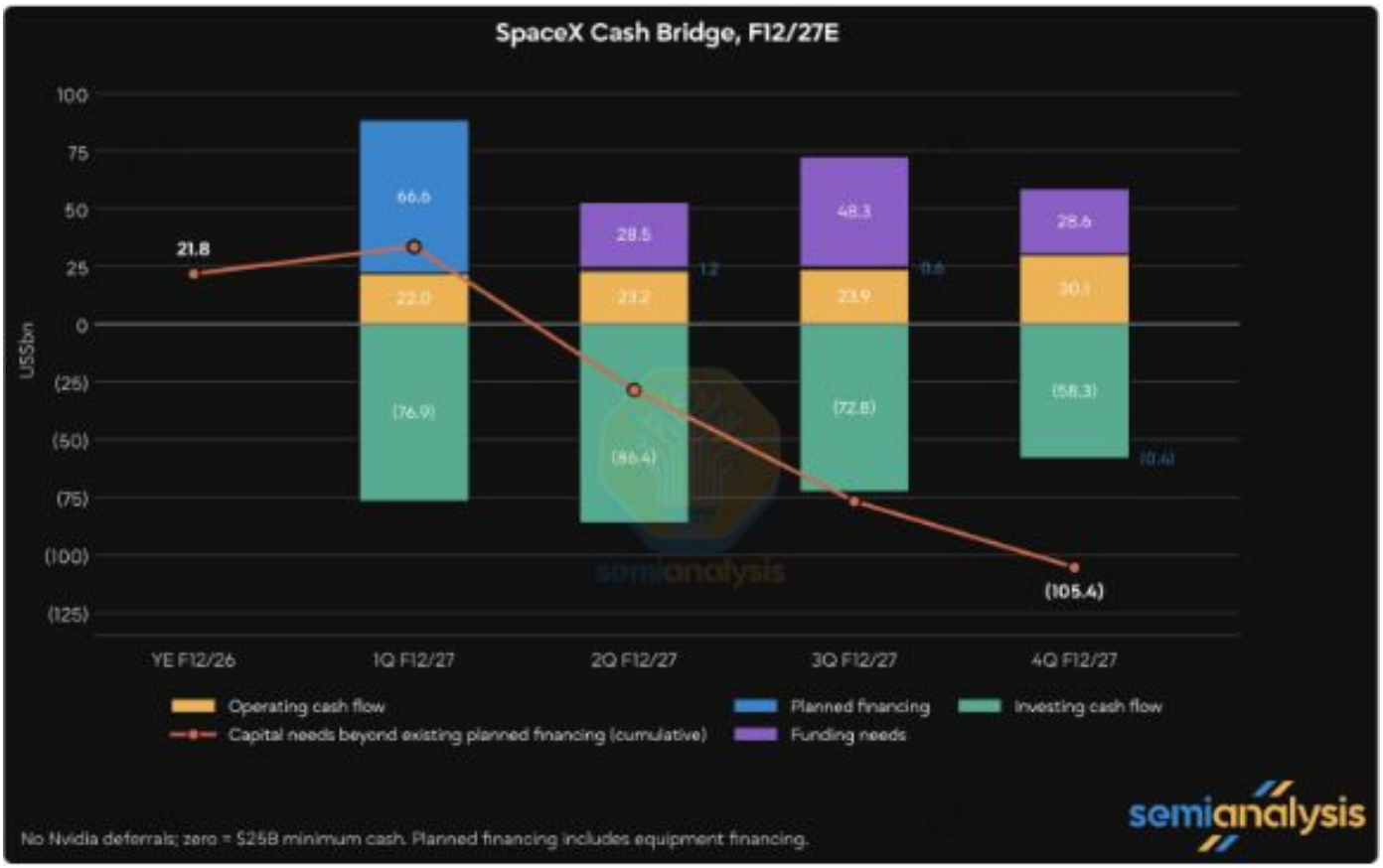
4Q F12/26. SpaceX starts with $78.9B of cash, of which $53.9B can be spent above the $25B minimum cash balance floor. Investing uses $48.6B against $14.9B of operating cash flow, before $0.6B of existing GPU principal payments. Equipment financing adds $2.1B. Cash ends the quarter at $46.8B, so no additional funding is required yet.

1Q F12/27. The proposed $40B unsecured debt issuance settles and the $25B Nvidia equity investment arrives, along with $2.3B of equipment financing. Operating cash flow contributes $22.0B against $76.9B of investing. After existing GPU and new equipment principal payments, cash ends at $58.5B. No additional funding is required this quarter.

2Q F12/27. Investing of $86.4B exceeds $23.2B of operating cash flow and $2.0B of equipment financing. After using cash above the floor and paying existing GPU and new equipment principal, SpaceX needs another $28.5B.

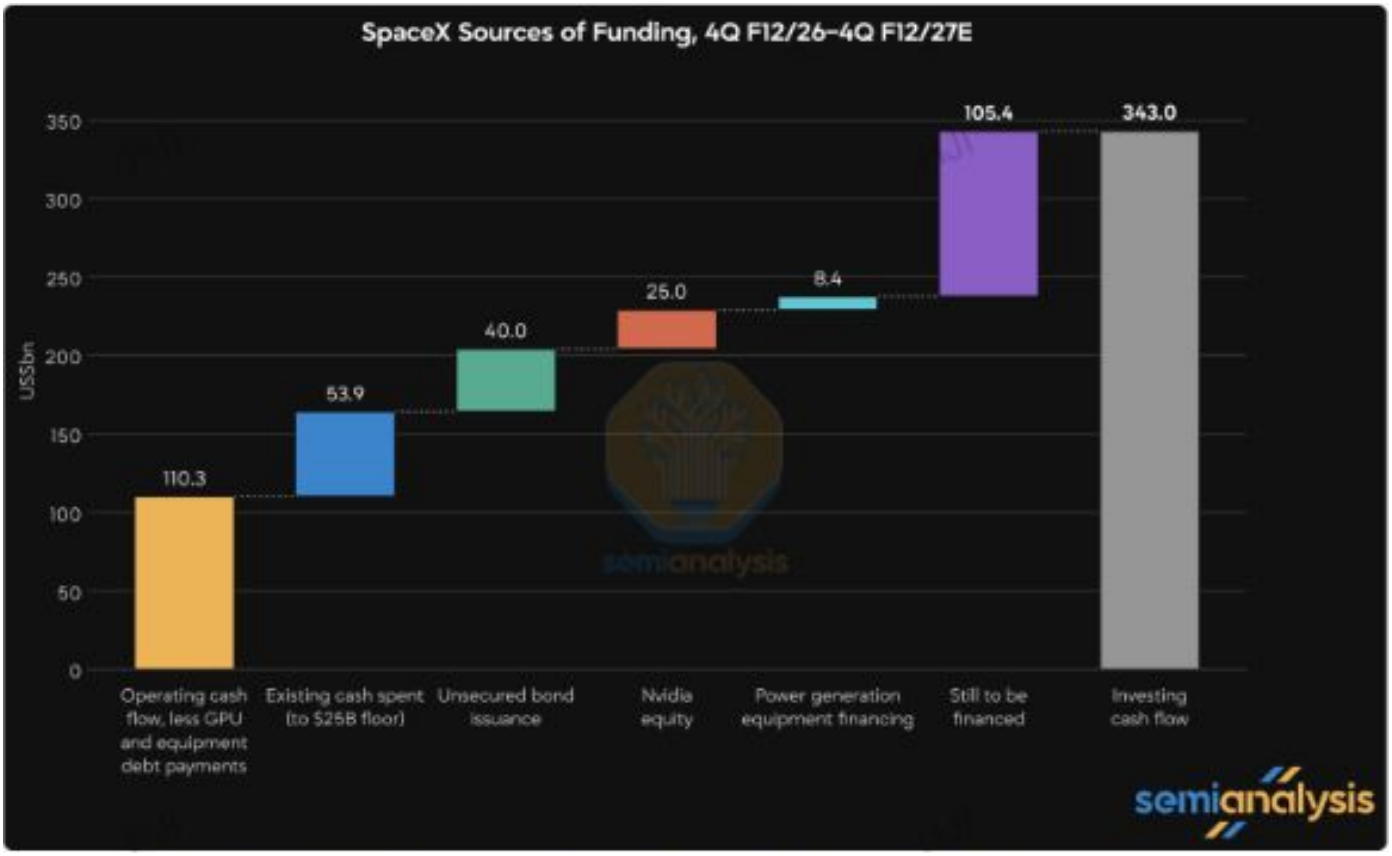
3Q F12/27. Investing eases to $72.8B against $23.9B of operating cash flow and $1.4B of equipment financing. Existing GPU and new equipment principal payments bring the remaining call to $48.3B. Cumulative additional funding reaches $76.8B.

4Q F12/27 onward. Operating cash flow of $30.1B remains below $58.3B of investing. After equipment financing and principal payments, another $28.6B brings cumulative additional funding to $105.4B. Committed site completions require $19.8B of AI cash capex in 2028. With no further discretionary expansion, quarterly operating cash flow reaches roughly $35B to $48B. If the gap is funded with a secured revolver (Tool A), operating cash retires it by 4Q F12/28. If the company opts for bonds instead, they remain outstanding.



Source: SemiAnalysis Compute Capital and Markets Model

Putting it all together, from 4Q F12/26 through 4Q F12/27, investing uses $343B. Operating cash after existing GPU and new equipment principal repayments contributes $110.3B, available cash resources contribute $53.9B, the most recent proposed unsecured debt issuance brings in $40B, Nvidia equity $25B and equipment financing $8.4B. That leaves $105.4B of residual capital to raise, which we assume carries an 8% cost of capital.



Source: SemiAnalysis Compute Capital and Markets Model

The scenario above forms our base case of instruments we think are highly likely to materialize, but how do we plug the remaining $105.4B of funding needs?

Five Tools to Fill The Gap – Three Traditional Debt Solutions and two Creative Solutions: Nvidia's "Buy Now Pay Later" Solution and Customer Prepayments

There are five tools at SpaceX's disposal: three traditional debt instruments (secured debt, unsecured bonds and GPU financing) and two less conventional sources, Nvidia's "Buy Now Pay Later" supplier credit and customer prepayments. They can serve different parts of the same financing plan.

Tool A: Secured Revolver Loan

A secured revolver can absorb uneven construction spending, equipment payments and customer collections. In the integrated case, SpaceX draws only when cash would fall below the $25B floor and repays as cash recovers. We assume a $40B limit and a 5.4% cost, comprising 3.9% SOFR and a 150 basis point spread. This bridge works alongside supplier credit, the proposed bond and loan package, equity and the separate bank facility.

Whether SpaceX can raise more secured debt will turn on collateral, existing pledges, permitted liens, guarantees and subsidiary restrictions. Taking on secured financing will also subordinate outstanding unsecured bonds, a clear negative, though this is mitigated by the fact that a revolver is meant to be short-term in nature.

Tool B: Buy Now and Pay Later (BNPL) – Nvidia Supplier Credit

Nvidia's 10-Q says that some investment grade customers can receive 90 days to a year to pay for large datacenter purchases. It does not identify SpaceX or disclose its credit limit, fees or security requirements.

Supplier credit connects funding to deliveries and gives installed GPUs time to earn before invoices mature. Its benefit depends on how quickly the equipment becomes billable and how much of the fleet produces cash. R&D and training earn no direct modeled revenue, and uncontracted capacity may not sell at the assumed rate.

Let's look at how much of the gap we can bridge just by using BNPL from Nvidia on a standalone basis, assuming one-year terms on direct Nvidia purchases delivered in 2H F12/26 through 3Q F12/27, with each quarter's invoices paid four quarters later.

Those invoices total $163.5B, which is also the peak supplier payable in 3Q F12/27: just four easy payments averaging $40.9B each to fund 2027's capex if SpaceX relies solely on Nvidia supplier credit. However, as the invoices mature, this case requires bridge financing later in 2028 as operating cash flow receipts don't come fast enough to pay back all of the supplier financing. SpaceX would have to draw on a revolver (possibly secured), with such borrowing peaking at $10.4B in 3Q F12/28. One-year invoices can, however, be paid by 3Q F12/28, and this bridge financing can be paid back by 4Q F12/28. We assume zero supplier fees as an illustration; this is not a verified offer.

What if we use BNPL together with the other tools? This would mean much less supplier financing: just $36.8B of temporary supplier credit on equipment delivered in 2H F12/26, of which $29.4B comes from Nvidia and $7.4B from other suppliers. Repayments are $14.7B in 1Q F12/27, $14.7B in 2Q F12/27 and $7.4B in 3Q F12/27.

Tool C: Unsecured Bonds

Unsecured term debt can fund a capital requirement that outlasts the shorter construction and supplier-credit bridges, and it has been the main tool SpaceX has been using to date. SpaceX's 5.35% notes due 2031 trade at T+150 over the 5y Treasury, a 6.52% yield. New bonds sized to be repaid within a few years should price at or a little wide of that, around T+175, or roughly 6.8% all-in. Our model assumes a more conservative 8% for unspecified additional capital. This represents a blend of further notes, converts and asset-level debt. At T+175, Tool C's net interest from 4Q F12/26 through 4Q F12/28 would be $17.6B rather than $19.2B. For comparison, SpaceX's existing $25B of notes carry a 5.855% weighted coupon, and our proxy for the proposed $40B issuance is T+300 over a 4.9% Treasury, or 7.9%.

The standalone case raises $105.4B of additional bonds, which only need to have a life of a few years (through 2028) to fund these capex needs, though we think SpaceX could seek to raise longer-dated debt. SpaceX would end 2028 with $151.7B of cash and $185.5B of debt. Paying 8% while cash earns an assumed 3.5% creates negative carry. A combined plan can reserve term funding for the requirement left after other, hopefully cheaper, tools are used. The model does not identify a committed issuance, establish demand at the assumed price or verify early redemption rights.

Tool D: Secured GPU Project Level Financing

GPU financing at the project or SPV level can fund eligible equipment and reduce cash paid at delivery. SpaceX reported $13.3B of related-party equipment debt, but unused borrowing capacity by cluster is not disclosed. The integrated base excludes new GPU financing. Taking out GPU financing at a 25% loan-to-cost ratio raises $68.7B against GPU and IT expenditure from 3Q F12/26 through 4Q F12/28, leaving $50.9B of that new financing outstanding at the end of 2028. Lender capacity, collateral, guarantees and borrowing restrictions remain conditional.

Secured GPU financing differs from Tool A in that Tool A is a corporate revolving facility that SpaceX draws and repays as cash moves, while GPU financing is asset-level debt tied to specific GPU deliveries, and it amortizes on a fixed schedule whether or not SpaceX has spare cash.

We assume an 11% cost, in line with what SpaceX pays on its Valor GPU financing – annualizing $327M of 2Q F12/26 related-party interest on about $11.9B of average principal gives ~11%. That rate is odd: secured, asset-level GPU debt should be the cheapest money SpaceX can raise yet, it costs 11% while the company's unsecured 2031 notes trade at T+150, a 6.52% yield. The likely explanation is recourse: GPU lenders look mainly to the chips, while bondholders lend against SpaceX as a whole. This includes Starlink, which we model at $30.4B of connectivity revenue in F12/27, a growing spectrum portfolio including the EchoStar purchase and Grain's 800 MHz licenses, and the equity value that comes with them. At roughly $2.1T of market capitalization, SpaceX's $39B of debt today is under 2% of its equity value, and even at the 4Q F12/27 peak in our Mix and

Match case, debt of about $154B is only ~7%, and by the end of F12/28 it is about 0.4x F12/28 EBITDA. On that measure SpaceX looks very underlevered, which helps explain why unsecured investors lend to it so cheaply. New GPU financing with a SpaceX parent guarantee should therefore price well inside 11%, and plausibly at or inside the unsecured curve. We keep 11% in the model, so Tool D's costs here are likely overstated.

However, issuing long-term or medium-term secured debt inserts a more senior obligation above existing bonds and will gravely sour unsecured debt investors' view of SpaceX and considerably hamper SpaceX's ability to access the market later.

Tool E: Customer Prepayments

We believe SpaceX can require customers to prepay part of a contract upfront to secure capacity. This would pull forward cash that would otherwise arrive later. Our base case assumes SpaceX sells its uncontracted capacity on one-year contracts with 30% of each year's contract value paid upfront. That brings in $20B of prepayments from 4Q F12/26 through 4Q F12/27 and $45.2B in aggregate through 2028. As the upfront payment will be netted off against billings over the one-y contract duration, net funding from customer prepayments would be smaller at ~$10B. On its own, Tool E can cut the $105.4B residual to $92.4B. In our Mix and Match case below, it cuts the residual from $23.2B to $10.4B.

The table below shows standalone sizing illustrations with different repayment rules. They do not establish available funding or an optimal mix. Tool A sweeps surplus cash, Tool B shows incremental payables, Tool C retains bonds, Tool D amortizes equipment debt and Tool E shows prepayments outstanding, leaving $92.4B to raise at 8%. Tool D uses a solved 49.3% GPU financing share. These balances cannot be added together.

SpaceX Standalone Financing Illustrations

Metric	Tool A	Tool B	Tool C	Tool D	Tool E
Instrument	Secured revolver	Nvidia BNPL (four-quarter terms)	Unsecured bonds	GPU financing	Customer prepayments
Forecast rate and service	5.4% drawn rate	Zero fee, 5.4% annual fee case	8% proxy	11% rate, Opening principal / 20 each quarter	No coupon, 30% of one-year contracts upfront
Peak obligation and quarter	$104.7B 4Q F12/27 Revolver principal	$149.0B 3Q F12/27 Incremental AP	$105.4B 4Q F12/27 New bonds	$123.5B 1Q F12/28 GPU / generation principal	$10.5B 3Q F12/28 Prepayments outstanding
Modeled balance reaches zero	4Q F12/28	Invoices: 3Q F12/28 Bridge: 4Q F12/28	Not by 4Q F12/28	Not by 4Q F12/28	Rolls over with renewals
Company net interest expense 4Q F12/26–4Q F12/28	$16.0B	$7.9B no fee $17.1B at 5.4%	$19.2B	$27.0B	$17.8B with the rest at 8%
Total debt at 4Q F12/28	$80.1B	$80.1B	$186.5B	$180.5B	$172.5B
Execution assumption	Collateral, covenants and pricing unverified	Eligible invoices and supplier approval. Bridge peaks at $10.4B in 3Q F12/28.	Market access and terms unverified	Solved share 49.3% of GPU/IT. Security and guarantees unverified.	$20.0B raised by 4Q F12/27. Leaves $92.4B to raise. Price unverified.

Standalone cases include the package, equity, generation financing and 30-day AP; integrated bank and credit features are off.
Liability peaks and repayment policies differ. Do not sum columns. Debt excludes Nvidia trade AP.
The 5.4% supplier fee adds $8.8B direct fees and $9.2B net interest. GPU principal remaining after 20 payments is 35.85%.
Tool E: 30% of each one-year contract on uncontracted capacity prepaid at the curve price; the remaining gap is raised at 8%.

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Source: SemiAnalysis Compute Capital and Markets Model. Tool B balances exclude ordinary 30-day payables.

Mix and Match – Our View on How SpaceX will Fund 2027 Capex

Our integrated case combines the instruments around the spending and collection schedule.

Temporary supplier credit covers $36.8B of eligible 4Q F12/26 equipment deliveries, and equipment financing funds half of on-site generation capex from 4Q F12/26.

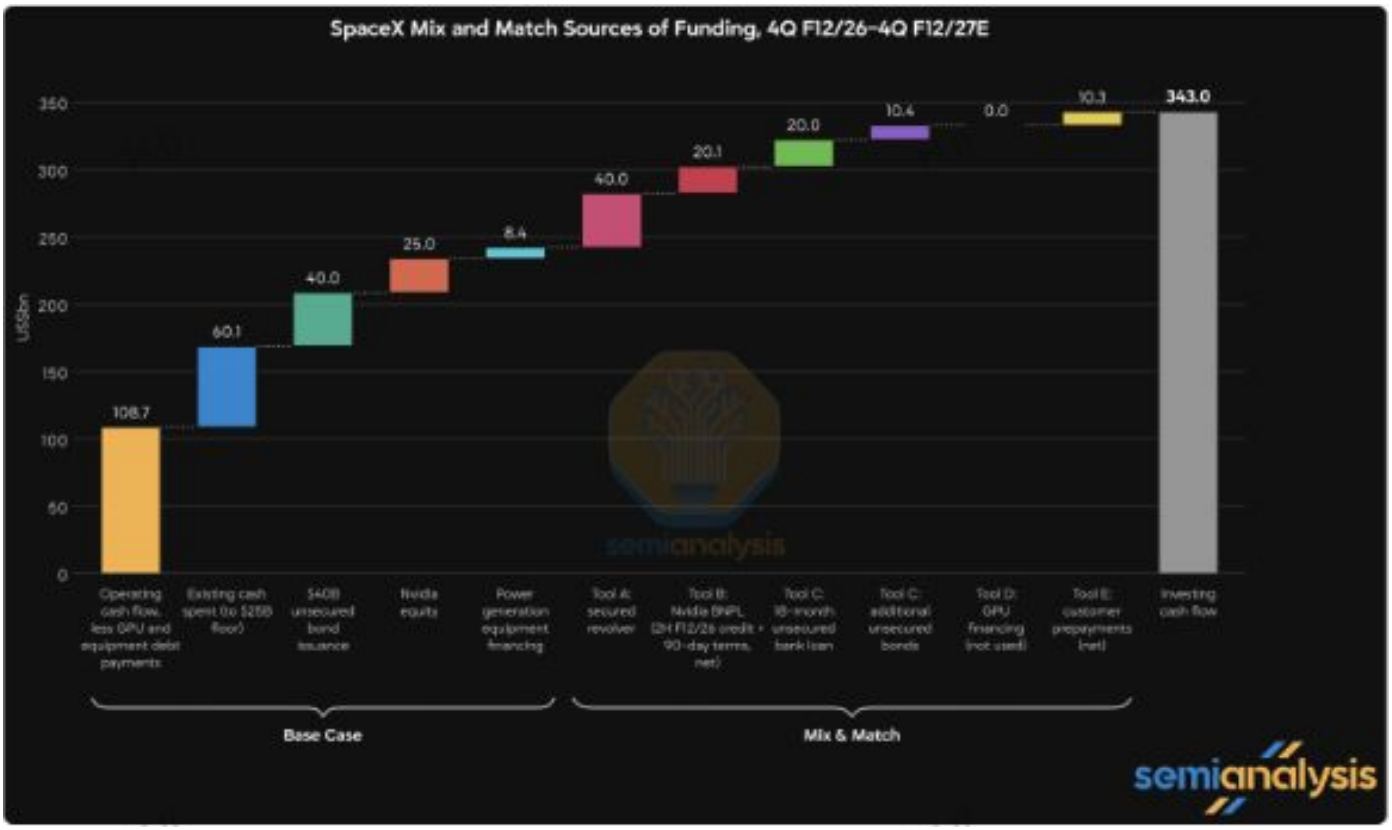
In 1Q F12/27, the proposed $40B bond and loan package and $25B Nvidia equity investment arrive as supplier repayments begin.

In 2Q F12/27, we model drawing an additional $20B unsecured 18-month bank facility. The $40B secured revolver is available from then but is first drawn in 3Q F12/27.

Supplier repayments continue through 3Q F12/27 while construction spending remains heavy. Ordinary 90-day Nvidia terms defer payment on separate eligible invoices, with that payable peaking at $51.0B in 2Q F12/27. The $40B secured revolver absorbs the remaining timing pressure, drawing $14.6B in 3Q F12/27, down from $26.3B without customer prepayments, which we assume at 30% of each one-year contract on uncontracted capacity.

By 4Q F12/27, the revolver hits its $40B limit, before the build is fully financed.

Even this combination requires another $10.4B in 4Q F12/27. The model raises this residual as unspecified additional capital at an assumed cost of capital of 8% to maintain the $25B cash floor. In practice we expect the unsecured piece to be split between the 18-month bank loan and about $10B of unsecured bonds, or termed out entirely in longer-dated bonds if SpaceX wants to build a war chest for 2028. We use no GPU financing (Tool D) in this case. The bridge below takes care of 2027. If SpaceX adds another 5 GW or more of capacity in 2028, we estimate roughly $150B or more of additional funding in 2028, raised with the same tools.



Source: SemiAnalysis Compute Capital and Markets Model.

Could SpaceX do better? Longer Nvidia BNPL terms shrink the residual requirement. With 180 days to pay instead of 90, SpaceX can rely on BNPL and the secured revolver, with usage of the revolver peaking at $17.8B

in 1Q F12/28 as invoices mature.

SpaceX stands to save a lot of money too. Net interest expense from 4Q F12/26 through 4Q F12/28 falls to $9.9B with 180 days to pay vs. $12.4B with 90 days. If SpaceX can manage to land 270 or 365 days to pay, neither additional capital nor the revolver is needed. Payables to Nvidia would peak at $139.9B and $170.2B, respectively. These are quarterly maturity illustrations; the main cash-flow sheet's payment-days sensitivity uses a different run-rate approximation.

SpaceX taking up Nvidia BNPL could be a win-win: if SpaceX can get free one-year payment terms, its total company net interest expense from 4Q F12/26 through 4Q F12/28 (interest on all of SpaceX's debt, including the existing notes and the $40B issuance, less interest earned on cash) would be $7.9B in our standalone comparison of each tool, against $15B with secured debt alone, $19.2B with unsecured bonds alone and $27B with GPU financing alone, a saving of $7.2B against the next cheapest option. Our Mix and Match case with 90-day terms and customer prepayments comes to $12.4B.

Nvidia also has a commercial incentive to finance equipment purchases, as we discussed in The Front End Gets Crowded. For Nvidia, offering financing terms enables a major client to build up compute at light speed, and the company is well positioned to understand the dynamics of SpaceX's short-term compute rental business and underwrite the risks involved.